Python

3 Months

Module 1

Python Environment Setup and Essentials

- Introduction to Python Language
- Features, the advantages of Python over other programming languages
- Python installation – Windows, Mac & Linux distribution for Anaconda Python
- Deploying Python IDE
- Basic Python commands, data types, variables, keywords and more

Module 2

Python language Basic Constructs

- Built-in data types in Python
- Learn classes, modules, Str(String), Ellipsis Object, Null Object, Ellipsis, Debug
- Basic operators, comparison, arithmetic, slicing and slice operator, logical, bitwise
- Loop and control statements while, for, if, break, else, continue.

Module 3

OOP concepts in Python

- How to write OOP concepts program in Python
- Connecting to a database
- Classes and objects in Python
- OOPs paradigm, important concepts in OOP like polymorphism, inheritance, encapsulation
- Python functions, return types and parameters
- Lambda expressions

Module 4

Database connection

- Understanding the Database, need of database
- Installing MySQL on windows
- Understanding Database connection using Python.

Module 5

NumPy for mathematical computing

- Introduction to arrays and matrices
- Broadcasting of array math, indexing of array
- Standard deviation, conditional probability, correlation and covariance.

Module 6

SciPy for scientific computing

- Introduction to SciPy
- Functions building on top of NumPy, cluster, linalg, signal, optimize, integrate, subpackages, SciPy with Bayes Theorem.

Module 7

Matplotlib for data visualization

- How to plot graph and chart with Python
- Various aspects of line, scatter, bar, histogram, 3D, the API of MatPlotLib, subplots.

Module 8

Pandas for data analysis and machine learning

- Introduction to Python dataframes
- Importing data from JSON, CSV, Excel, SQL database, NumPy array to dataframe
- Various data operations like selecting, filtering, sorting, viewing, joining, combining

Module 9

Exception Handling

- Introduction to Exception Handling
- Scenarios in Exception Handling with its execution
- Arithmetic exception
- RAISE of Exception
- What is Random List, running a Random list on Jupyter Notebook
- Value Error in Exception Handling.

Module 10

Multi Threading & Race Condition

- Introduction to Thread, need of threads
- What are thread functions
- Performing various operations on thread like joining a thread, starting a thread, enumeration in a thread
- Creating a Multithread, finishing the multithreads.
- Understanding Race Condition, lock and Synchronization.

Module 11

Packages and Functions

- Intro to modules in Python, ne
- How to import modules in py

ogram Curriculum

- Locating a module, namespace and scoping
- Arithmetic operations on Modules using a function
- Intro to Search path, Global and local functions, filter functions
- Python Packages, import in packages, various ways of accessing the packages
- Decorators, Pointer assignments, and Xldr.

Module 12

Web scraping with Python

- Introduction to web scraping in Python
- Installing of beautifulsoup
- Installing Python parser lxml
- Various web scraping libraries, beautifulsoup, Scrapy Python packages
- Creating soup object with input HTML
- Searching of tree, full or partial parsing, output print

Skills to Master

- **Python Basic Constructs**
- **Multi-threading**
- **Packages & Functions**
- **Exception Handling**
- **Python for Apache Spark**
- **Database Connections**
- **Matplotlib**
- **Pandas NumPy & SciPy**
- **OOPS in Python**
- **Web Scraping**